FDTT 704: Advanced Statistics

Dr. Mutua Kilai, PhD

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Department of Pure and Applied Sciences

Course Purpose

This course is to equip researchers in fashion design with advanced statistical, data management, and visualization skills necessary for conducting empirical research, analyzing trends, and presenting findings effectively.

Expected Learning Outcomes

By the end of the course, students should be able to:

i. Use descriptive and inferential statistical methods to analyze fashion-related data.

ii. Collect, clean, and structure data efficiently using software like SPSS, R, Python, and SQL.

Cont’d

iii. Apply statistical and analytical techniques to optimize garment production, quality, and fashion marketing strategies.

iv. Assess the validity and reliability of quantitative research methods in fashion studies.

Course Content

Role of data in fashion research and design decision-making. Tools for data management: SQL, Excel, Google Sheets, and database systems. Research Ethics in Fashion Data Management; Inferential statistics (t-tests, chi-square, ANOVA, correlation, regression); hypothesis testing, correlation, and regression analysis. Factor analysis, principal component analysis (PCA), and cluster analysis. Software Tools for Data Visualization: Tableau, Power BI, and Python libraries (Matplotlib, Seaborn). Qualitative data analysis; Understanding market trends using statistical modeling. Psychometric Analysis in Fashion Research.

Teaching Methodologies

Lecture, Independent Research, Interactive Seminars and Colloquia, Guest Lectures & Expert Panels, One-on-one mentorship and supervision from faculty experts, Research, Team-Based Problem Solving, class discussions and presentations.

Instructional Materials/Equipment

Fashion journals, digital and online learning platforms, Industry Reports & Fashion Analytics, case study analysis, thesis/dissertations, conference proceedings, laptop computer, and LCD projector

Course Assessment

Assignment,Discussion,Presentations	**10%**
Practical/Case Analysis	10%
Written Continuous Assessment Tasks	20%
Final Examination	60%

> **Important**
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> The passmark is 50% but of **essence is to ensure you have got the skills**

About Me

- A Data Science and Statistics Lecturer at Kirinyaga University, Department of Pure and Applied Sciences
- Passionate about using data to make impact in real life.
- PhD in Statistics, MSc. Applied Statistics, BSc. Statistics

Software

🚧 Caution

- Since we only have 14 weeks of lecture, we only concentrate on one software for data analysis; R.
- Ensure you download R and RStudio latest version.
- We will do an intridinctory session on R

Role of data in fashion research and design decision-making

Introduction

- Data is abundant in the current fashion business environment as sales data, product information, and consumer data are constantly collected and analyzed.
- The importance of data has been gradually acknowledged by fashion professionals to improve sales and margins because fashion brands and retailers need to develop, manufacture, and sell styles that resonate with consumers.
- Lately, with advancements in data analytics and machine learning, fashion brands and retailers have become more aware of the value of utilizing Artificial Intelligence (AI)-based software or applications to create efficient fashion design, merchandising, and marketing strategies.

Cont’d

- The most significant reason fashion brands and retailers are seeking the ways to optimize data analytics is to improve their product offerings through precise personalization for targeted customers, which would eventually bring higher sell-through and profitability.

- The availability of machine learning algorithms, big data, and cheap but high-powered computing has brought significant changes in many industries including fashion by providing meaningful insights from various data sources.

- Fashion brands and retailers started making use of their internal data to increase sales and profitability and to remain competitive in the market.

Cont’d

- Gruzbarg proposed the four dimensions to make “behavioral and contextual targeting” possible by using data analytics as follows:

1. Demographics like gender, age, or income to understand who the customer is

2. Historic/behavioral data including purchase transactions, cart abandoned, etc. to understand what the customer did in the past and can be used to predict their future behavior

3. Streaming/contextual data such as a customer’s current digital behaviors, web pages they are viewing, emails they just open ads they are clicking on to know in what product the customer is interested and whether s/he is currently in shopping mode

4. Predictive analytics using AI-powered software to predict how the customer is likely to respond in the future by identifying invisible patterns in the previous three dimensions

Strategic Decision Making in Fashion Industry Using Data Analytics

Product Design

Data analytics helps designers identify consumer preferences, colours, fabrics and styles that are trending, enabling the development of products that better satisfy customer needs.

Inventory Management

Historical sales data can be analysed to forecast demand, reducing stock shortages and excess inventory while improving supply chain efficiency.

Customer Segmentation

Statistical techniques such as clustering can group customers according to purchasing behaviour, demographics or preferences, allowing personalized marketing strategies.

Pricing Decisions

Regression models and market analyses help determine the relationship between price and demand, enabling firms to optimize pricing strategies and maximize profits.

Tools for data management: SQL, Excel, Google Sheets, and database systems.

Introduction

- Data refers to raw facts, figures, observations, images, measurements or recordings collected during research or business operations.
- Examples in Fashion industry include; customer ages, garment sizes, fabric prices, colour preferences, textile test results, sales records, fashion trend reports, production costs among others.

> **Data Management**
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> Data management is the process of collecting, storing, organizing, cleaning, protecting, retrieving and sharing data for decision making.

Cont’d

- Good data management ensures that information is:
  i. Accurate
  ii. Complete
  iii. Secure
  iv. Accessible
  v. Reliable

Common Data Management Tools

- The mostly widely used tools include:

  a. Microsoft Excel

  b. Google Sheets

  c. SQL

  d. Database Management Systems

Microsoft Excel

- Microsoft Excel is a spreadsheet application used to organize, analyze and visualize data.
- It stores information in rows, columns and worksheets
- Each file is called a workbook

Common Excel Features

Tables, sorting, filtering, charts, pivot tables, conditional formating, formulas, functions, data validation

Advantages of Excel

- Easy to learn.
- Widely available.
- Good for small datasets.
- Supports charts and graphs.
- Excellent for descriptive statistics.
- Can import data from many sources.
- Suitable for preliminary analysis.

Limitations

- Difficult to manage millions of records.
- Multiple users cannot edit efficiently.
- High risk of accidental deletion.
- Version control problems.
- Weak security for sensitive information.

Google Sheets

- Google Sheets is a cloud-based spreadsheet developed by Google.
- Unlike Excel, files are stored online.
- Multiple users can work on the same document simultaneously.

Features

Online storage, Automatic saving, Real-time collaboration,Sharing permissions,Version history,Integration with Google Forms

Advantages

- Free with Google account
- Accessible anywhere
- Supports teamwork
- Automatic backup
- No software installation
- Easy sharing
- Works on mobile devices

Limitations

- Requires internet for best performance
- Slower with very large datasets
- Fewer advanced statistical tools than Excel
- Limited database capabilities

SQL (Structured Query Language)

- It is a programming language used to: create databases, store data, retrieve data, update data, delete data.

- When data become too large for Excel, SQL becomes more efficient.

Advantages

- Handles very large datasets
- Fast retrieval
- High security
- Reduces duplication
- Supports multiple users
- Easy data integration
- Reliable

Disadvantages

- Requires learning SQL commands
- More technical than spreadsheets
- Requires database software
- Initial setup may be complex

Database Management Systems (DBMS)

- A database is an organized collection of related information stored electronically.
- A Database Management System is software used to; create databases, manage databases, secure databases, retrieve information, backup data.
- Examples include: MySQL, PostgreSQL,Oracle Database, Microsoft SQL Server,SQLite

Advantages

- Very secure
- Handles millions of records
- Reduces redundancy
- Multiple users
- Fast searching
- Automatic backup
- Supports business intelligence

Limitation

- Expensive to implement
- Requires database administrators
- Technical expertise needed
- More complex than spreadsheets

Best Practices for Data Management in Fashion Research

Cont’d

- Control access using passwords and user permissions.
- Validate data entry to minimize errors.
- Comply with institutional and legal guidelines on data privacy and ethics.

Research Ethics in Fashion Data Management

Introduction to Research Ethics

- Research ethics refers to the moral principles and professional standards that guide researchers in conducting research responsibly, honestly, and respectfully.

Cont'd

- Research ethics ensures that:

   i. Participants are treated fairly

   ii. Information is collected responsibly

   iii. Privacy is protected

   iv. Research findings are reported honestly

   v. Society benefits from research

Ethical Data Management

- Ethical data management is the responsible handling of research data throughout its lifecycle.

- This includes: planning data collection, obtaining permission, collecting data, storing data securely,analyzing data honestly,sharing data responsibly,preserving data safely.

Ethical Principles in Fashion Research

Respect for Persons

- Researchers should respect the dignity, rights, and autonomy of every participant.
- Participants should: decide voluntarily whether to participate, understand the purpose of the research, withdraw at any time without penalty.

> **Example**
>
> A researcher studying customer satisfaction in luxury fashion stores should clearly explain the study before asking customers to complete a questionnaire.

Beneficence

- Researchers should maximize benefits while minimizing harm.
- Possible harms include: emotional discomfort, financial loss, invasion of privacy, reputational damage.

Example

Publishing confidential information about a fashion company's production costs may harm the company's competitiveness.

Justice

- All participants should be treated fairly.
- Researchers should avoid selecting participants based solely on convenience if it leads to unfair representation.

> **Example**
>
> A study on sustainable fashion should include consumers from different age groups, income levels, and regions rather than focusing only on urban shoppers.

Integrity

- Researchers should be honest throughout the research process.
- Integrity includes: truthful reporting, accurate analysis, proper citation, avoiding fabrication, avoiding falsification, avoiding plagiarism.

Best Practices in Ethical Fashion Data Management

- obtain ethical approval before starting the study,
- secure informed consent,
- collect only necessary data,
- anonymize personal information,
- store data securely,
- back up files regularly,

Cont’d

- maintain accurate records,
- report findings honestly,
- cite all sources appropriately,
- comply with institutional and legal regulations,
- dispose of data securely when no longer required.

Introduction to R Statistical Software

Contents

⚠ **What You Will Learn**

- Use R and RStudio interfaces
- Run R commands
- Create R objects
- Write your own R functions and scripts
- Load and use R packages
- Create quick plots